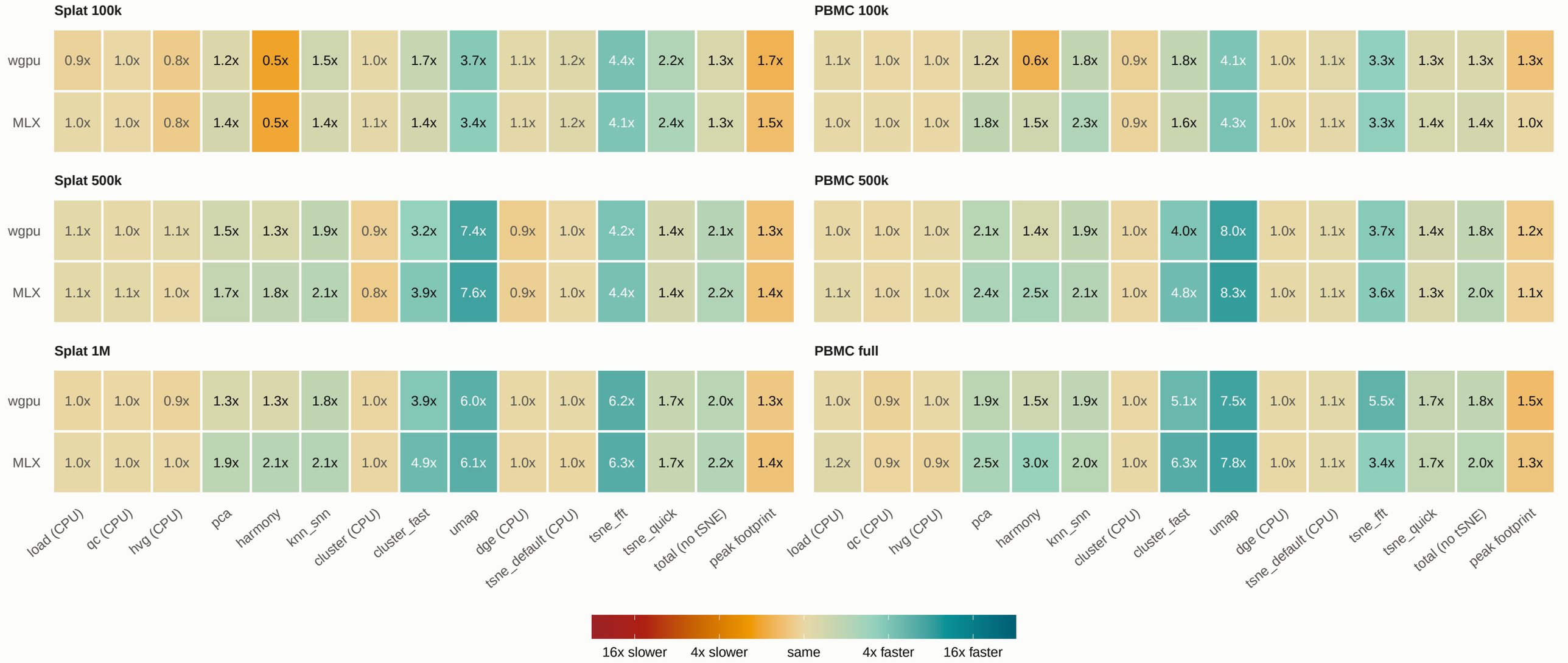
Native GPU against native CPU, per step

Speed−up of bixverse.gpu (wgpu, MLX) over the same bixverse pipeline on the CPU, both native on macOS. One repeat each.



(CPU): the same CPU code in every row, so these tiles show run−to−run noise. Default tSNE runs the Barnes−Hut optimiser on the CPU.
Peak footprint: GPU over CPU peak (macOS /usr/bin/time −l, whole run), coloured red when the GPU row needs more.
(qd) steps tsne_quick and cluster_fast are not in the total. M1 Max, 8 threads, R 4.6.1.